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KNN

Part of **Future Connect Media's** IT Course

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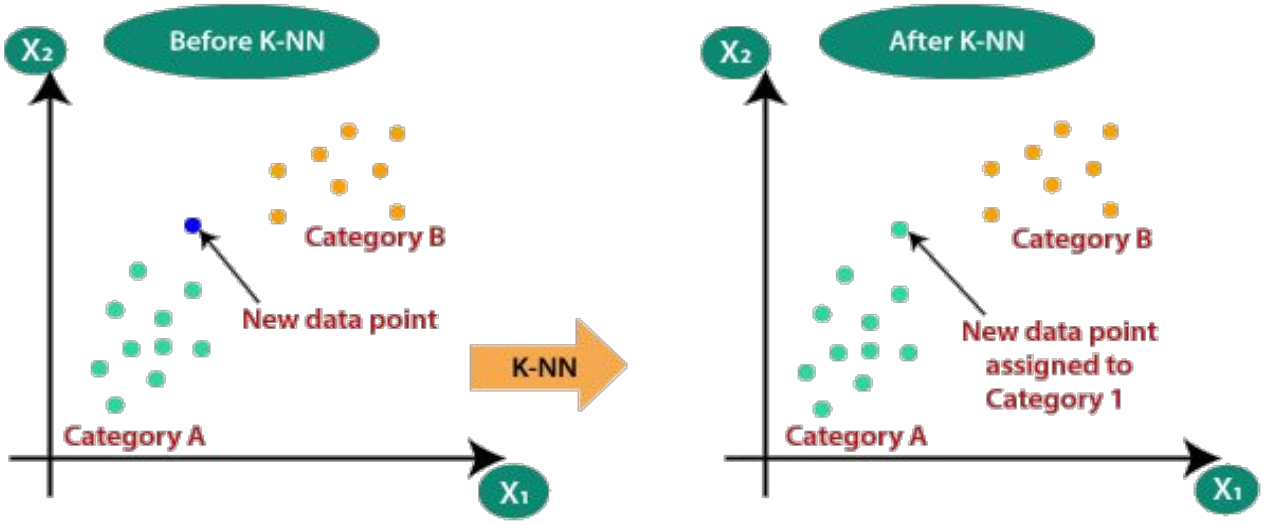
Topics to be covered:

- What is KNN?
- How we Made it?

KNN

Definition: K-Nearest Neighbors (KNN) is a simple, non-parametric supervised learning algorithm used for classification and regression tasks.

Basic Principle: KNN classifies data points based on the classes of their nearest neighbors in the feature space. The class of the majority of the K-nearest neighbors determines the class of the new data point.



How KNN Works

1. **Choose K Value:** Select the number of neighbors (K) to consider for making predictions. A common starting point is K = 3 or K = 5.
2. **Calculate Distances:** For a given data point that needs classification, calculate the distance between this point and all other points in the training dataset using the chosen distance metric.
3. **Identify Neighbors:** Sort the calculated distances and identify the K-nearest neighbors.
4. **Voting Mechanism (for Classification):**
   - Each of the K neighbors votes for their class.
   - The class with the majority votes becomes the predicted class for the new data point.
5. **Averaging (for Regression):**
   - For regression tasks, the predicted value is the average of the values of the K-nearest neighbors.

Key Concept of KNN

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Supervised Learning: KNN is a type of supervised learning where the model is trained on labeled data, meaning that the input features are associated with output labels (classes).

Distance Metrics:

- **Euclidean Distance:** The most commonly used distance metric. Calculated as:

$$d(x, y) = \sqrt{\sum_{i=1}^{n} (x_i - y_i)^2}$$

- **Manhattan Distance:** Another option, calculated as:

$$d(x, y) = \sum_{i=1}^{n} |x_i - y_i|$$

- **Minkowski Distance:** Generalizes both Euclidean and Manhattan distances.

Choosing Right K

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Impact of K:

- **Small K Value:** May lead to overfitting as the model becomes too sensitive to noise in the training data.
- **Large K Value:** May lead to underfitting, where the model is too generalized and may miss local patterns.

Techniques for Choosing K:

- **Cross-Validation:** Use cross-validation to determine the optimal K value based on model performance metrics (accuracy, F1 score, etc.).
- **Elbow Method:** Plot the error rate versus different values of K and look for an "elbow" point where the error stabilizes.